BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: C Batch: 2 Practical No: 10

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Subject Name: Deep Learning

Problem Statement:

Design and implement deep learning application for audio classification that incorporates Speech to Text and Text to Speech using appropriate deep learning architecture and evaluate its performance using suitable metrics.

Algorithm:

1. Start the program and install the required libraries.
2. Import the necessary libraries and load the Whisper deep learning model.
3. Upload the input audio file in WAV, MP3, or M4A format.
4. Read the audio file and preprocess it for speech recognition.
5. Pass the audio to the Whisper model to recognize spoken words.
6. Convert the recognized speech into text and display the transcription.
7. Enter the correct reference transcript and calculate Word Error Rate (WER).
8. Pass the recognized text to the gTTS engine.
9. Generate the speech audio file, play the output, and download it.
10. Display the results, evaluate performance, and stop the program.

Code:

```
# ============================================================
# DEEP LEARNING PRACTICAL NO. 10
# Audio Processing: Speech-to-Text and Text-to-Speech
# ============================================================

# CELL 1: Install required libraries
!pip -q install openai-whisper gTTS jiwer
!apt-get -qq update
!apt-get -qq install -y ffmpeg

print("Required libraries installed successfully!")


# CELL 2: Import libraries
import os
```

```
import time
import whisper

from gtts import gTTS
from jiwer import wer
from google.colab import files
from IPython.display import Audio, display, Markdown

print("Libraries imported successfully!")


# CELL 3: Load Whisper deep learning model
model = whisper.load_model("base")

print("Whisper model loaded successfully!")


# CELL 4: Upload an audio file
print("Upload an audio file (WAV, MP3, or M4A):")

uploaded = files.upload()

audio_file = next(iter(uploaded.keys()))

print("Uploaded file:", audio_file)


# CELL 5: Speech-to-Text
start_time = time.time()

result = model.transcribe(audio_file)

stt_time = time.time() - start_time

recognized_text = result["text"].strip()
detected_language = result.get("language", "unknown")

print("\n========== SPEECH-TO-TEXT RESULTS ==========")
print("Input audio:", audio_file)
print("Recognized text:", recognized_text)
print("Detected language:", detected_language)
print(f"Processing time: {stt_time:.2f} seconds")

print("\nOriginal audio:")
display(Audio(filename=audio_file))


# CELL 6: Evaluate Speech-to-Text using WER
reference_text = input(
    "Enter the correct transcript of the uploaded audio: "
).strip()
```

```
if reference_text and recognized_text:
    error_rate = wer(
        reference_text.lower(),
        recognized_text.lower()
    )

    print("\n========== STT EVALUATION ==========")
    print("Reference transcript:", reference_text)
    print("Recognized transcript:", recognized_text)
    print(f"Word Error Rate (WER): {error_rate:.4f}")
    print(f"Word Error Rate: {error_rate * 100:.2f}%")

    if error_rate == 0:
        print("Result: Exact word match.")
    else:
        print("Result: Transcription contains word errors.")

    if error_rate <= 1:
        print(f"Approximate word accuracy: {(1-error_rate)*100:.2f}%")
    else:
        print("Word accuracy: Not reported because WER exceeds 100%.")
else:
    error_rate = None
    print("Reference or recognized text is empty. Evaluation skipped.")


# CELL 7: Text-to-Speech
print("\n========== TEXT-TO-SPEECH ==========")

if recognized_text:
    tts_start_time = time.time()

    tts = gTTS(
        text=recognized_text,
        lang=detected_language
        if detected_language in ["en", "hi", "fr", "es", "de", "it",
"pt"]
        else "en"
    )

    output_file = "/content/generated_speech.mp3"
    tts.save(output_file)

    tts_time = time.time() - tts_start_time

    print("Speech generated successfully!")
    print(f"TTS generation time: {tts_time:.2f} seconds")

    print("\nGenerated speech:")
    display(Audio(filename=output_file))
```

```
    files.download(output_file)
else:
    tts_time = 0
    print("No recognized text available for speech generation.")


# CELL 8: Display recognized text
print("\n========== RECOGNIZED TEXT ==========")

display(Markdown("### Speech-to-Text Output"))
display(Markdown(recognized_text if recognized_text else "No text
recognized."))


# CELL 9: Final performance evaluation
print("\n" + "=" * 50)
print("FINAL PROJECT RESULTS")
print("=" * 50)

print("Input audio file:", audio_file)
print("Detected language:", detected_language)
print("Recognized text:", recognized_text)
print(f"STT processing time: {stt_time:.2f} seconds")
print(f"TTS generation time: {tts_time:.2f} seconds")

if error_rate is not None:
    print(f"Word Error Rate: {error_rate * 100:.2f}%")

    if error_rate <= 1:
        print(f"Approximate word accuracy: {(1-error_rate)*100:.2f}%")

if os.path.exists("/content/generated_speech.mp3"):
    print("TTS output: Generated successfully")
    print("Output format: MP3")
else:
    print("TTS output: Not generated")

print("=" * 50)
print("Program completed successfully!")
```

OUTPUT:

```
Required libraries installed successfully!
Libraries imported successfully!
100%|████████████████████████████████████████| 139M/139M [00:01<00:00, 128MiB/s]
Whisper model loaded successfully!
Upload an audio file (WAV, MP3, or M4A):
 Choose Files  sample_en...speech.wav
sample_english_speech.wav(audio/wav) - 793232 bytes, last modified: 10/20/2026 - 100% done
Saving sample_english_speech.wav to sample_english_speech.wav
Uploaded file: sample_english_speech.wav
/usr/local/lib/python3.13/dist-packages/whisper/transcribe.py:132: UserWarning: FP16 is not supported on CPU; using FP32 instead
  warnings.warn("FP16 is not supported on CPU; using FP32 instead")
```

```
========== SPEECH-TO-TEXT RESULTS ==========
Input audio: sample_english_speech.wav
Recognized text: Hello, my name is Alex. Today I am talking about technology and artificial intelligence. Deep learning helps computers recognize speech, understand language, and make useful pred
Detected language: en
Processing time: 7.91 seconds

Original audio:
```

▶ 0:00 / 0:17

```
Enter the correct transcript of the uploaded audio: Hello, my name is Alex. Today I am talking about technology and artificial intelligence. Deep learning helps computers recognize speech, unders

========== STT EVALUATION ==========
Reference transcript: Hello, my name is Alex. Today I am talking about technology and artificial intelligence. Deep learning helps computers recognize speech, understand language, and make useful
Recognized transcript: Hello, my name is Alex. Today I am talking about technology and artificial intelligence. Deep learning helps computers recognize speech, understand language, and make usefu
Word Error Rate (WER): 0.0000
Word Error Rate: 0.00%
Result: Exact word match.
Approximate word accuracy: 100.00%

========== TEXT-TO-SPEECH ==========
Speech generated successfully!
TTS generation time: 1.85 seconds

Generated speech:
```

▶ 0:00 / 0:18

```
========== RECOGNIZED TEXT ==========
```

Speech-to-Text Output

Hello, my name is Alex. Today I am talking about technology and artificial intelligence. Deep learning helps computers recognize speech, understand language, and make useful predictions. This audio is a sample for a speech recognition practical.

```
==================================================
FINAL PROJECT RESULTS
==================================================
Input audio file: sample_english_speech.wav
Detected language: en
Recognized text: Hello, my name is Alex. Today I am talking about technology and artificial intelligence. Deep learning helps computers recognize speech, understand language, and make useful pred
STT processing time: 7.91 seconds
TTS generation time: 1.85 seconds
Word Error Rate: 0.00%
Approximate word accuracy: 100.00%
TTS output: Generated successfully
Output format: MP3
==================================================
Program completed successfully!
```